

* Atmega 128

1. Features :

- i) High Performance , Low Power , AVR 8-bit Microcontroller
- ii) Advanced RISC Architecture
 - 133 Powerful Instruction
 - 32 x 8 Purpose Register peripheral control Reg
 - Fully Static operation
 - UP TO 16 MIPS Throughput at 16MHz
 - only 8 bit 2-cycle multiplier
- iii) High Endurance non volatile memory
 - 128 KBytes , self programmable Flash memory
 - 4 KBytes EEPROM
 - 4 KBytes internal SRAM
 - write / Erase cycles: 10,000 / 100,000 EEPROM
 - UP to 64 KBytes optional External memory
 - SPI interface for In-system Programming

Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri
14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30				

2026

* Peripheral features

- Two 8-Bit Timer/Counter with separate prescalers and Compare Modes
- Two Expanded 16-Bit Timer compare mode & Capture Mode
- Real Time Counter with separate oscillator
- Two 8-bit PWM channels
- 6 PWM channels with Programmable Resolution from 2 to 16 Bits
- 10-Bit, 8 channels ADC
- Byte-oriented Two-wire serial interface
- Dual Serial USARTs
- Master / slave SPI

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14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	1	2	3

Monday

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I/O and Package

- 53 Programmable I/O Lines
- 64-lead TQFP and 64-Pad QFN

operating voltage

- 2.7 → 5.5 V ATmega128 (L)
- 4.5 → 5.5 V ATmega128

Speed Grades

- 0 → 8 MHz ATmega128 (L)
- 0 → 16 MHz ATmega128

November

Sunday

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DIO Peripheral

* ATmega 32

→ have 32 Pins

→ For Ports → Port (A, B, C, D)

→ every port have completed 8 Bit

→ Addresses From 0x20 → 0x5F

→ have internal Pull up for every port

Saturday

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نوفمبر

السبت

٤ جمادي الآخر ١٤٤٨ هـ

* ATmega 128

→ have 53 Pins

→ 7 Ports → Port (A, B, C, D, E, F, G)

→ The port from A → F have 8 Bit
except (Port G) only 5 Bit

→ Addresses from 0x20 → 0x6F

→ have Internal Pull up for every 53 Pins

* in function of Pins we write

if (PinNumber < 4) in Group G
so we check before work on Bits

* in function of Groups we write
Group G & 0x1F to protect Pins 5, 6, 7

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		1	2	3	4	5	6	7	8	9	10	11	12
				24	25	26	27	28	29	30			

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EXT Peripheral :* ATmega 32

- have 3 interrupts (INT0, INT1, INT2)
- have 2 registers → MCUCR (INT0, INT1)
8 MCUCSR for (INT2)
- have GICR (Global interrupt control reg)
- have GIFR (Global interrupt control Flag)
- have 3 pins PD2 → INT0
PD3 → INT1
PB2 → INT2
- INT0 & INT1 support 4 modes
INT2 support only Edge mode
- ISR Vector

vector 1

vector 2

vector 3

* ATmega 128

→ have 8 Interrupt (INT0 → INT7)

→ have 2 Register → EICRA (INT0 → INT3)
 & EICRB (INT4 → INT7)

→ EIMSK [External Interrupt Mask Register]
 → EIFR [External Interrupt Flag Register]

→ INT0 → PD0	, INT4 → PE4
INT1 → PD1	, INT5 → PE5
INT2 → PD2	, INT6 → PE6
INT3 → PD3	, INT7 → PE7

→ All 8 Interrupt Support 4 Modes

→ ISR vector — vector: 1 → — vector: 8

ADC Driver

* ATmega 32

→ have PortA → pins PA0 → PA7

→ 8 channels → ADC0 → ADC7

→ limited differential channel Gain (1x, 2x, 4x, 8x)

→ have 5 Registers → ADMUX, ADCSRA, ADCH, ADCL, SFIOR

→ ISR vector → vector 16

2026

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* ATmega 128

- have PORT F → Pins (PF0 → PF7)
- have 8 channels (ADC0 → ADC7)
- Expanded differential channel (1x, 10x, 20x)
- have 5 registers → ADMUX
ADCSRA, ADCH, ADCL, SFIQR
- ISR vector → vector 21